## 6.5.4 Forces and motion

### 6.5.4.1 Describing motion along a line

### 6.5.4.1.1 Distance and displacement

| Content | Key opportunities for skills development |
|---|---|
| Distance is how far an object moves. Distance does not involve direction. Distance is a scalar quantity.<br><br>Displacement includes both the distance an object moves, measured in a straight line from the start point to the finish point and the direction of that straight line. Displacement is a vector quantity.<br><br>Students should be able to express a displacement in terms of both the magnitude and direction. | MS 1, 3c<br><br>Throughout this section (Forces and motion), students should be able to use ratios and proportional reasoning to convert units and to compute rates. |

### 6.5.4.1.2 Speed

| Content | Key opportunities for skills development |
|---|---|
| Speed does not involve direction. Speed is a scalar quantity.<br><br>The speed of a moving object is rarely constant. When people walk, run or travel in a car their speed is constantly changing.<br><br>The speed at which a person can walk, run or cycle depends on many factors including: age, terrain, fitness and distance travelled.<br><br>Typical values may be taken as:<br><br>walking ~ 1.5 m/s<br><br>running ~ 3 m/s<br><br>cycling ~ 6 m/s.<br><br>Students should be able to recall typical values of speed for a person walking, running and cycling as well as the typical values of speed for different types of transportation systems.<br><br>It is not only moving objects that have varying speed. The speed of sound and the speed of the wind also vary.<br><br>A typical value for the speed of sound in air is 330 m/s. | |
| Students should be able to make measurements of distance and time and then calculate speeds of objects.<br><br>For an object moving at constant speed the distance travelled in a specific time can be calculated using the equation: | MS 1a, c, 2f |

| Content | Key opportunities for skills development |
|---|---|
| $distance\ travelled = speed \times time$<br>$[\ s = v\,t\ ]$<br>distance, $s$, in metres, m<br>speed, $v$, in metres per second, m/s<br>time, $t$, in seconds, s | MS 3b, 3c<br>Students should be able to recall and apply this equation. |
| Students should be able to calculate average speed for non-uniform motion. | MS 3b, 3c |

### 6.5.4.1.3 Velocity

| Content | Key opportunities for skills development |
|---|---|
| The velocity of an object is its speed in a given direction. Velocity is a vector quantity.<br>Students should be able to explain the vector–scalar distinction as it applies to displacement, distance, velocity and speed.<br>(HT only) Students should be able to explain qualitatively, with examples, that motion in a circle involves constant speed but changing velocity. | |

### 6.5.4.1.4 The distance–time relationship

| Content | Key opportunities for skills development |
|---|---|
| If an object moves along a straight line, the distance travelled can be represented by a distance–time graph.<br>The speed of an object can be calculated from the gradient of its distance–time graph.<br>(HT only) If an object is accelerating, its speed at any particular time can be determined by drawing a tangent and measuring the gradient of the distance–time graph at that time.<br>Students should be able to draw distance–time graphs from measurements and extract and interpret lines and slopes of distance–time graphs, translating information between graphical and numerical form.<br>Students should be able to determine speed from a distance–time graph. | MS 4a, b, c, d, f |

### 6.5.4.1.5 Acceleration

| Content | Key opportunities for skills development |
|---|---|
| The average acceleration of an object can be calculated using the equation: | |
| $acceleration = \frac{change\ in\ velocity}{time\ taken}$<br>$\left[\ a = \frac{\Delta v}{t}\ \right]$<br>acceleration, *a*, in metres per second squared, m/s$^2$<br>change in velocity, $\Delta v$, in metres per second, m/s<br>time, *t*, in seconds, s<br>An object that slows down is decelerating.<br>Students should be able to estimate the magnitude of everyday accelerations. | MS 1d, 3b, 3c<br>Students should be able to recall and apply this equation. |
| The acceleration of an object can be calculated from the gradient of a velocity–time graph.<br>(HT only) The distance travelled by an object (or displacement of an object) can be calculated from the area under a velocity–time graph.<br>Students should be able to: | MS 4a, b, c, d, f |
| • draw velocity–time graphs from measurements and interpret lines and slopes to determine acceleration<br>• (HT only) interpret enclosed areas in velocity–time graphs to determine distance travelled (or displacement) | WS 3.3 |
| • (HT only) measure, when appropriate, the area under a velocity–time graph by counting squares. | WS 3.3 |
| The following equation applies to uniform acceleration:<br>$(final\ velocity)^2 - (initial\ velocity)^2 = 2 \times acceleration \times distance$<br>$\left[\ v^2 - u^2 = 2\ a\ s\ \right]$<br>final velocity, *v*, in metres per second, m/s<br>initial velocity, *u*, in metres per second, m/s<br>acceleration, *a*, in metres per second squared, m/s$^2$<br>distance, *s*, in metres, m<br>Near the Earth's surface any object falling freely under gravity has an acceleration of about 9.8 m/s$^2$. | MS 3b, 3c<br>Students should be able to apply this equation which is given on the *Physics equation sheet*. |

| Content | Key opportunities for skills development |
|---|---|
| An object falling through a fluid initially accelerates due to the force of gravity. Eventually the resultant force will be zero and the object will move at its terminal velocity. | |

## 6.5.4.2 Forces, accelerations and Newton's Laws of motion

### 6.5.4.2.1 Newton's First Law

| Content | Key opportunities for skills development |
|---|---|
| Newton's First Law:<br><br>If the resultant force acting on an object is zero and:<br><br>• the object is stationary, the object remains stationary<br>• the object is moving, the object continues to move at the same speed and in the same direction. So the object continues to move at the same velocity.<br><br>So, when a vehicle travels at a steady speed the resistive forces balance the driving force.<br><br>So, the velocity (speed and/or direction) of an object will only change if a resultant force is acting on the object.<br><br>Students should be able to apply Newton's First Law to explain the motion of objects moving with a uniform velocity and objects where the speed and/or direction changes.<br><br>(HT only) The tendency of objects to continue in their state of rest or of uniform motion is called inertia. | |

### 6.5.4.2.2 Newton's Second Law

| Content | Key opportunities for skills development |
|---|---|
| Newton's Second Law: | |
| The acceleration of an object is proportional to the resultant force acting on the object, and inversely proportional to the mass of the object.<br><br>As an equation: | MS 3a<br><br>Students should recognise and be able to use the symbol for proportionality, $\propto$ |

| Content | Key opportunities for skills development |
|---|---|
| $resultant\ force = mass \times acceleration$<br>$F = m\,a$<br>force, *F*, in newtons, N<br>mass, *m*, in kilograms, kg<br>acceleration, *a*, in metres per second squared, $m/s^2$ | MS 3b, c<br>WS 4.2<br>Students should be able to recall and apply this equation. |
| (HT only) Students should be able to explain that:<br>• inertial mass is a measure of how difficult it is to change the velocity of an object<br>• inertial mass is defined as the ratio of force over acceleration. | MS 3a |
| Students should be able to estimate the speed, accelerations and forces involved in large accelerations for everyday road transport.<br>Students should recognise and be able to use the symbol that indicates an approximate value or approximate answer, ~ | MS 1d |

**Required practical activity 19:** investigate the effect of varying the force on the acceleration of an object of constant mass, and the effect of varying the mass of an object on the acceleration produced by a constant force.

AT skills covered by this practical activity: physics AT 1, 2 and 3.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in Key opportunities for skills development (page 189).

#### 6.5.4.2.3 Newton's Third Law

| Content | Key opportunities for skills development |
|---|---|
| Newton’s Third Law: | |
| Whenever two objects interact, the forces they exert on each other are equal and opposite.<br>Students should be able to apply Newton’s Third Law to examples of equilibrium situations. | WS 1.2 |

### 6.5.4.3 Forces and braking

### 6.5.4.3.1 Stopping distance

| Content | Key opportunities for skills development |
|---|---|
| The stopping distance of a vehicle is the sum of the distance the vehicle travels during the driver's reaction time (thinking distance) and the distance it travels under the braking force (braking distance). For a given braking force the greater the speed of the vehicle, the greater the stopping distance. | |

### 6.5.4.3.2 Reaction time

| Content | Key opportunities for skills development |
|---|---|
| Reaction times vary from person to person. Typical values range from 0.2 s to 0.9 s.<br>A driver's reaction time can be affected by tiredness, drugs and alcohol. Distractions may also affect a driver's ability to react.<br>Students should be able to:<br>• explain methods used to measure human reaction times and recall typical results | |
| • interpret and evaluate measurements from simple methods to measure the different reaction times of students | WS 3.5, 3.7 |
| • evaluate the effect of various factors on thinking distance based on given data. | WS 1.5, 2.2<br>MS 1a, c<br>AT 1<br>Measure the effect of distractions on reaction time. |

### 6.5.4.3.3 Factors affecting braking distance 1

| Content | Key opportunities for skills development |
|---|---|
| The braking distance of a vehicle can be affected by adverse road and weather conditions and poor condition of the vehicle.<br>Adverse road conditions include wet or icy conditions. Poor condition of the vehicle is limited to the vehicle's brakes or tyres.<br>Students should be able to:<br>• explain the factors which affect the distance required for road transport vehicles to come to rest in emergencies, and the implications for safety | |

| Content | Key opportunities for skills development |
|---|---|
| • estimate how the distance required for road vehicles to stop in an emergency varies over a range of typical speeds. | MS 1c, 1d, 2c, 2d, 2f, 2h, 3b, 3c |

### 6.5.4.3.4 Factors affecting braking distance 2

| Content | Key opportunities for skills development |
|---|---|
| When a force is applied to the brakes of a vehicle, work done by the friction force between the brakes and the wheel reduces the kinetic energy of the vehicle and the temperature of the brakes increases.<br>The greater the speed of a vehicle the greater the braking force needed to stop the vehicle in a certain distance.<br>The greater the braking force the greater the deceleration of the vehicle. Large decelerations may lead to brakes overheating and/or loss of control.<br>Students should be able to: | |
| • explain the dangers caused by large decelerations | WS 1.5 |
| • (HT only) estimate the forces involved in the deceleration of road vehicles in typical situations on a public road. | (HT only) MS 1d |